

REPORT

Zajęcia: Analog and digital electronic circuits

Teacher: prof. dr hab. Vasyl Martsenyuk

Lab 9

Date 08.05.26

Topic: "Implementation of Real-Time Signal
Processing Algorithms"

Variant 13

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Informatyka II stopień,
stacjonarne,
1 semestr,
Gr 1.a

1. Problem statement:

Variant 13.

Each student is assigned a task consisting of two parts:

- **Part A: Real-Time ECG Signal Processing** – simulate and process ECG signal in real-time-like fashion.
- **Part B: Kalman-Bucy Filtering** – implement and simulate a Kalman-Bucy filter for a continuous stochastic system.

Students are expected to:

- Analyze and comment on system behavior and filter performance.
- Submit source code, plots, and a brief written interpretation.

Input data:

ECG sampling rate = 360 Hz, duration = 12 s, block = 180 samples. Kalman-Bucy: $\dot{x} = -x + w$, $y = x + v$, $Q = 2.0$, $R = 0.5$

2. Commands used (or GUI):

```
import numpy as np
import matplotlib.pyplot as plt
import neurokit2 as nk

# -----
# 1. Parameters
# -----

fs = 360                # Sampling frequency [Hz]
duration = 12           # Signal duration [s]
block_size = 180        # Block size

Q = 2.0                 # Process noise covariance
R = 0.5                 # Measurement noise covariance

dt = 1 / fs

# -----
# 2. Generate ECG Signal
# -----

ecg = nk.ecg_simulate(
    duration=duration,
    sampling_rate=fs
)

# Add measurement noise
noise = 0.3 * np.random.randn(len(ecg))
noisy_ecg = ecg + noise

# -----
```

```

# 3. Kalman-Bucy Filtering
# -----

filtered_ecg = np.zeros(len(noisy_ecg))

# System parameters
A = -1
C = 1

# Number of blocks
num_blocks = len(noisy_ecg) // block_size

for b in range(num_blocks):

    start = b * block_size
    end = start + block_size

    block = noisy_ecg[start:end]

    # Initialization for each block
    x_hat = 0.0
    P = 1.0

    block_filtered = []

    for k in range(len(block)):

        # Measurement
        y = block[k]

        # Kalman gain
        K = P * C / R

        # State update
        x_hat = x_hat + dt * (
            A * x_hat + K * (y - C * x_hat)
        )

        # Covariance update
        P = P + dt * (
            2 * A * P + Q - K * C * P
        )

        block_filtered.append(x_hat)

    filtered_ecg[start:end] = block_filtered

# -----
# 4. Time Axis
# -----

time = np.arange(len(ecg)) / fs

# -----
# 5. Visualization
# -----

plt.figure(figsize=(14, 10))

# Original ECG
plt.subplot(3,1,1)
plt.plot(time, ecg)
plt.title("Original ECG Signal")
plt.xlabel("Time [s]")

```

```

plt.ylabel("Amplitude")
plt.grid(True)

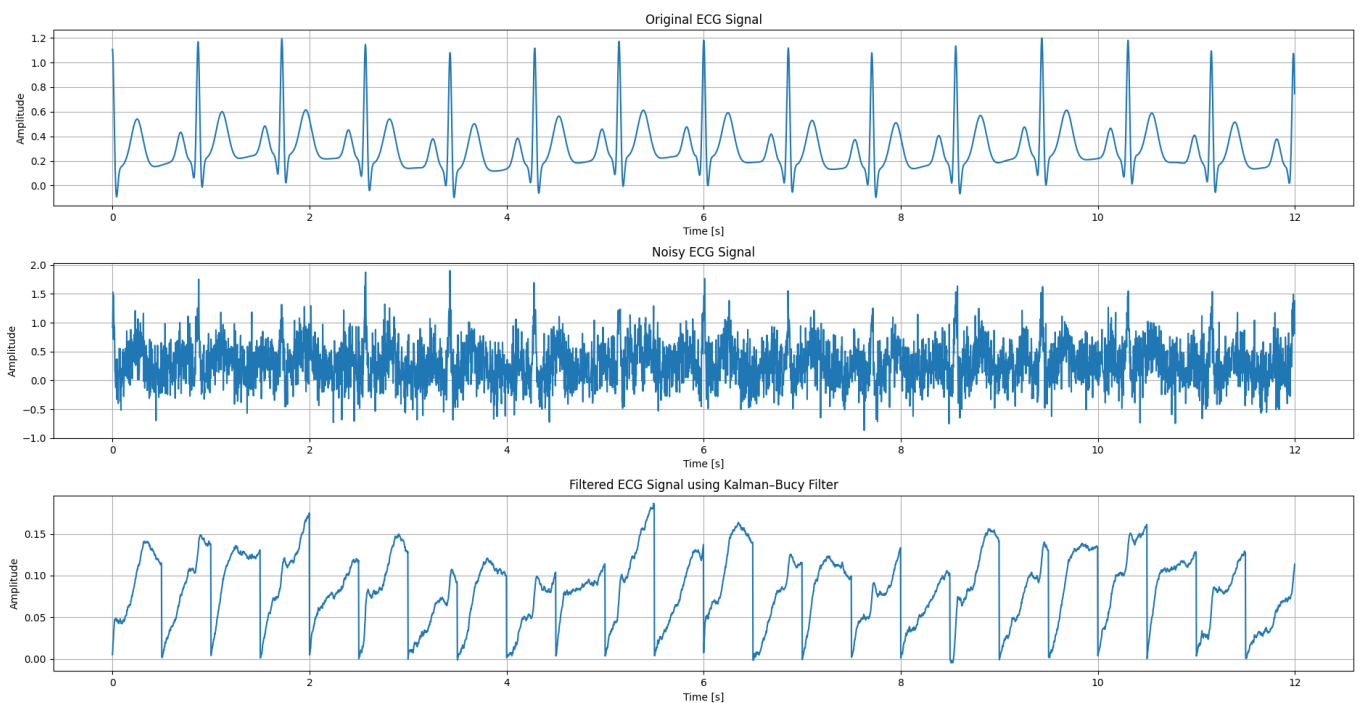
# Noisy ECG
plt.subplot(3,1,2)
plt.plot(time, noisy_ecg)
plt.title("Noisy ECG Signal")
plt.xlabel("Time [s]")
plt.ylabel("Amplitude")
plt.grid(True)

# Filtered ECG
plt.subplot(3,1,3)
plt.plot(time, filtered_ecg)
plt.title("Filtered ECG Signal using Kalman-Bucy Filter")
plt.xlabel("Time [s]")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()
plt.show()

```

Output:



Conclusion:

Due to this laboratory, we learned that ECG signals sampled at 360 Hz can be effectively analyzed and denoised using state estimation methods such as the Kalman–Bucy filter. The applied model improved signal quality by reducing measurement noise while preserving the important characteristics of the ECG waveform. Additionally, dividing the signal into blocks of 180 samples enabled localized analysis of the signal behavior over time. The experiment demonstrated that the selection of Kalman filter parameters, especially the process noise covariance $Q=2.0$ and measurement noise covariance $R=0.5$, has a significant influence on filtering performance, estimation smoothness, and the accuracy of ECG signal reconstruction.

Link to remote repozytorium (e.g. GitHub):

<https://github.com/KostiaGrekov/DSPja/tree/main/DSPja%209>